

HINTS AND SOLUTIONS

Q.1 A

Sol. Inside the metallic ball, potential remain constant

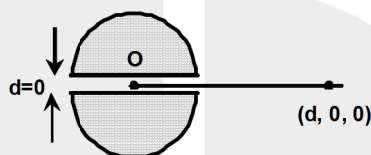
$$\therefore \text{potential at G} = \text{potential at C} = \frac{Q}{4\pi\epsilon_0 L}$$

Q.2 D

Q.3 A

Sol. Due to a solid hemispherical charge of $2Q$ the field will be $\frac{1}{4\pi\epsilon_0} \times \frac{2Q}{d^2}$ along x-axis. So due to one hemisphere the component of field along x-direction will be $\frac{1}{4\pi\epsilon_0} \times \frac{Q}{d^2}$. And hence the net field will be

$$\text{more than } \frac{1}{4\pi\epsilon_0} \times \frac{Q}{d^2}.$$



Q.4 D

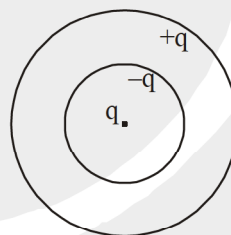
Q.5 C

Q.6 A

Q.7 C

$$\text{Sol. } u = \frac{k(q)(-q)}{a} + \frac{kqq}{b} + \frac{kq(-q)}{b} + \frac{k(-q)^2}{2a} + \frac{kq^2}{2b}$$

$$u = \frac{kq^2}{2b} - \frac{kq^2}{2a}$$



Q.8 B

Q.9 D

Q.10 A

Q.11 C

$$\text{Sol.(i) } s_y = 1\text{cm} = \frac{1}{2} \times \frac{qE}{m} t^2$$

$$s_x = 10\text{ cm} = v_x t = 300t$$

$$t = \frac{1}{30} \text{ sec.}$$

$$\text{(ii) } v_y^2 = 2a_y s_y$$

$$v_y = 15 \times 4 \times 10^{-2} = 6 \times 10^{-1} \text{ m/s}$$

- Q.12 D
Q.13 BC
Q.14 BC

Sol. Let the points charges have the coordinate

$$P \equiv (-a/2, 0, 0), Q \equiv (a/2, 0, 0) \text{ and } R \equiv (0, \frac{\sqrt{3}}{2}a, 0)$$

$\therefore$ coordinate of point at which E has to be calculated is $O \equiv (0, \frac{a}{2\sqrt{3}}, \sqrt{\frac{2}{3}}a)$.

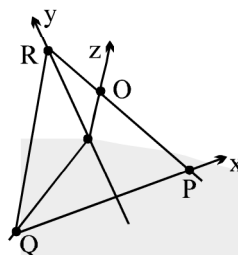
Also we have

$$\vec{E} = \frac{1}{4\pi\epsilon_0} \frac{Q}{r^3} \vec{r}$$

$$\therefore \vec{E}_{\text{net}} = \vec{E}_1 + \vec{E}_2 + \vec{E}_3$$

$$\therefore \vec{E}_{\text{net}} = \frac{1}{4\pi\epsilon_0} \frac{Q}{a^3} (\vec{P}_0 + \vec{Q}_0 + \vec{R}_0)$$

$$= \frac{Q}{4\pi\epsilon_0 a^3} \cdot \left(3\sqrt{\frac{3}{2}}a \right) \hat{k} = \frac{\sqrt{6}Q}{4\pi\epsilon_0 a^2} \hat{k}$$



Q.15 AB

Q.16 [Ans. (A) – S; (B) – P; (C) – R; (D) – Q]

Sol. (A) $E_x = \left(-\frac{dv}{dx} \right) \hat{i} = -1\hat{i}$
 $E_y = \left(-\frac{dv}{dx} \right) \hat{j} = \frac{1}{\sqrt{3}} \hat{j}$

$$\tan\theta = \frac{1}{\sqrt{3}} = 30^\circ$$

$$\theta = 150^\circ$$

(B) $E_x = \left(-\frac{dv}{dx} \right) \hat{i} = +1\hat{i}$

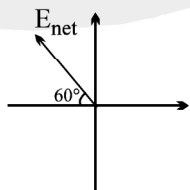
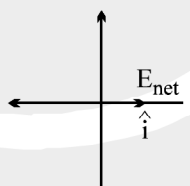
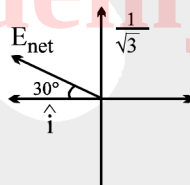
$$E_y = \left(-\frac{dv}{dx} \right) \hat{j} = 0$$

$$\theta = 0^\circ$$

(C) $E_x = \left(-\frac{dv}{dx} \right) \hat{i} = -1\hat{i}$

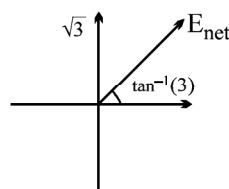
$$E_y = \left(-\frac{dv}{dx} \right) \hat{j} = +\sqrt{3}\hat{j}$$

$$\theta = 120^\circ$$



$$(D) \quad E_x = \left(-\frac{dv}{dx}\right)\hat{i} = -\left(-\frac{1}{\sqrt{3}}\right)\hat{i} = \frac{1}{\sqrt{3}}\hat{i}$$

$$E_y = \left(-\frac{dv}{dy}\right)\hat{j} = -(-\sqrt{3})\hat{j} = \sqrt{3}\hat{j} \quad \text{Ans.}$$



Alternative

$$\vec{E} = -\frac{\delta V}{\delta x} - \frac{\delta V}{\delta y} \hat{j}$$

$$(A) \quad \vec{E} = -\hat{i} + \frac{1}{\sqrt{3}}\hat{j} \quad \theta = 150^\circ$$

$$(B) \quad \vec{E} = +\hat{i} + 0\hat{j} \quad \theta = 0^\circ$$

$$(C) \quad \vec{E} = -\hat{i} + \sqrt{3}\hat{j} \quad \theta = 120^\circ$$

$$(D) \quad \vec{E} = \frac{1}{\sqrt{3}}\hat{i} + \sqrt{3}\hat{j} \quad \theta = \tan^{-1}(3) \quad 60^\circ < \theta < 90^\circ$$

$$Q.17 \quad [\text{Ans. } Q\sqrt{\frac{K}{10RM}}]$$

$$Q.18 \quad \text{Ans (a)} \quad U_2 = \frac{a^2 Q^2}{8\pi\epsilon_0 R} \left(\frac{1-a^n}{1-a}\right)^2 \quad \text{where } a = \frac{R}{r+R}, \quad (b) \quad U_2 (n \rightarrow \infty) = \frac{RQ^2}{8\pi\epsilon_0 r^2}$$

$$Q.19 \quad [\text{Ans. } 0.234]$$

$$\text{Sol.} \quad E_{\text{disc}} = \frac{\sigma}{2\epsilon_0} \left(1 - \frac{x}{\sqrt{x^2 + R^2}}\right); \quad \sigma = \frac{Q}{A} = \frac{Q}{\pi R^2}$$

$$E_{\text{ring}} = \frac{KQx}{(x^2 + R^2)^{3/2}}$$

$$\text{Put } X = 3\text{m}, R = 4\text{m}, \epsilon_0 = \frac{1}{4\pi K}$$

$$Q = 1 \times 10^{-6} \text{ C}, K = 9 \times 10^9 \text{ Nm}^2/\text{C}^2$$

$$E_{\text{Net}} = 234 \text{ N/C}$$

$$Q.20 \quad [\text{Ans. } (\sigma/2\epsilon_0)(\sin\theta_2 - \sin\theta_1)]$$

$$Q.21 \quad [\text{Ans. } 6 \text{ m}]$$

$$\text{Sol.} \quad \frac{\lambda}{2\pi\epsilon_0 r} = E_{\text{break}}$$

$$r = \frac{\lambda}{2\pi\epsilon_0 E_{\text{break}}}$$

$$= \frac{10^{-3}}{2 \times 3.14 \times 8.85 \times 10^{-12} \times 3 \times 10^6}$$

$$= \frac{1}{2 \times 3.14 \times 8.85} \times 10^3$$

$$= 5.99 \text{ m} \approx 6 \text{ m} \quad \text{Ans.}$$

